RAJ BHARAT RASAL

AI Systems & Multi-Agent Infrastructure Engineer

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PROFESSIONAL SUMMARY

AI Systems & Backend Architect specializing in multi-agent orchestration, decoupled mediation topologies, and LLM intent routing. Engineered F.R.I.D.A.Y (3-agent decoupled backend) and EduOS curiosity engine, emphasizing ground-truth verification over unverified generation. Adept at constructing reliable, auditable agent pipelines with zero circular dependencies.

TECHNICAL & ARCHITECTURAL SKILLS

Languages: Python, TypeScript / JavaScript, SQL (PostgreSQL, SQLite)

Distributed Systems & Architecture: Decoupled Orchestration, System Design & Scalability

AI & Machine Learning: Multi-Agent AI Pipelines, LLM Intent Classification & Routing, Prompt Engineering & Chains, RAG & Ground Truth Verification, Adversarial Critic Systems

Backend & Cloud: FastAPI, RESTful API Design, Asynchronous Programming (asyncio), APScheduler & Task Queues

Security & Systems: Sandboxing & Isolation (Firecracker)

Soft Skills: Technical Architecture Documentation, Conceptual Simplification, Problem Decomposition

FEATURED SYSTEMS & PROJECTS

SATURDAY / SUNDAY — Architect & Lead Developer

Jan 2026 - Present

Tech Stack: Python, C++, Tensor Sharding, Zero-Trust Protocols, LAN, Tailscale VPN, Socket Programming

- Built a distributed compute network to run large local LLMs, solving hardware limitations of processing complex AI models on single devices.
- Implemented automatic hardware discovery and dynamic tensor sharding to split a single model's processing across heterogeneous Mac and Windows nodes via LAN and Tailscale VPN mesh.
- Designed a zero-trust communication protocol between nodes, utilizing proof-of-compute verification to detect and reject nodes returning corrupted or falsified matrix calculation results.
- **Key Architecture:** Decoupled coordinator-worker model with heartbeats; distributed matrix multiplication with Tailscale mesh tunnels; sub-millisecond node health checks; Byzantine fault tolerance through spot verification.

F.R.I.D.A.Y — Multi-Agent Communication Backend — Lead Backend Architect & Developer

Jun 2026 - Present

Tech Stack: Python, FastAPI, Baileys API (WhatsApp), APScheduler, asyncio, LLM Intent Routing

- Originated at SummerHacks 2026 in a team of 4; currently driving continuous backend evolution and multi-agent infrastructure.
- Designed a strictly decoupled 3-agent architecture (Herald, Nexus, Dispatch) enforcing complete separation of concerns so agents never call each other directly, eliminating circular deadlocks.
- Implemented LLM intent classification with confidence-based routing logic to catch ambiguous queries and trigger clarification flows before dispatching execution tasks.
- Engineered resilient background job scheduling and automated task pipelines with robust fault recovery.
- **Key Architecture:** 3-tier mediator topology; zero direct agent coupling; confidence threshold routing (<0.85 triggers conversational disambiguation); asynchronous message queue via FastAPI.

EduOS — AI Curiosity-First Education Engine — Creator & Lead Developer

Nov 2025 - Feb 2026

Tech Stack: Python, FastAPI, WhatsApp Cloud API, Prompt Chains, Knowledge Graph, SQLite

- Developed an AI education platform featuring an algorithmic 'pinch engine' that generates one high-value conceptual knowledge connection per day for students based on their curiosity graph.
- Engineered core logic prioritizing curiosity-driven concept delivery before standard syllabus notes, boosting daily student engagement metrics.
- Built WhatsApp API integration, deliberately decoupled during development with an isolated rate-limiter and queue to prevent unauthorized broadcasting.
- **Key Architecture:** Curiosity graph mapping; asynchronous outbound dispatch with token bucket rate limiting; deterministic prompt chain templates with ground-truth validation.

EXPERIENCE & LEADERSHIP

Independent Tutoring Practice — Mathematics & Physics Tutor (Grades 10–12)

Jun 2024 - Present

- Taught 50+ high school students, translating complex scientific, mathematical, and algorithmic concepts into clear, structured lessons.

- Developed diagnostic assessment frameworks to identify conceptual bottlenecks and tailored individual curricula to improve performance by an average of 22%.
- Honed mastery of first-principles thinking, communication clarity, and pedagogical system design.

SYSTEM DESIGN SPECIFICATIONS & WHITEPAPERS

HAMIA Architecture Specification: Hierarchical Adaptive Multi-Model Intelligence Architecture (Architecture Whitepaper, Oct 2026)

- A cloud-native fusion system that orchestrates a federation of specialized frontier models (12+) through an 8-stage pipeline. Emphasizes evidence-centric fusion over naive voting, using external ground-truth verifiers (Firecracker sandboxes, SymPy) and a cyclic adversarial critic to prevent multi-model ensembles from amplifying shared bias.

TWELVE Product Documentation: AI-Powered Viva & Interview Assessment System (System Design & Product Specification, Oct 2026)

- A secure, kiosk-based assessment pipeline that replaces human examiner bias with a multi-model AI panel. Features a specialized split-evaluator architecture (Curriculum, Application, Coherence) and a robust legal/compliance strategy for deployment in higher education environments.

EDUCATION

ITM Skills University — Bachelor of Technology (B.Tech)

Aug 2025 - May 2029

Field: Computer Science Engineering | Score: Undergraduate (In Progress)

- Specializing in Distributed Systems, Operating Systems, Computer Networks, and AI Systems Design. Active contributor to student engineering and hackathon teams.

Central Board of Secondary Education (CBSE) — Higher Secondary Certificate (12th Grade)

2023 - 2025

Field: Science (PCM) | Score: 83.4%

- Strong foundation in advanced calculus, mechanics, electromagnetic physics, and algorithmic logic.

Central Board of Secondary Education (CBSE) — Secondary School Examination (10th Grade)

2021 - 2023

Field: General Science & Mathematics | Score: 96.0%

- Academic excellence award in Mathematics and Science.

ACHIEVEMENTS & SECURITY RESEARCH

- **3rd Place — Deal Dome Competition:** Awarded 3rd place for designing the architectural specification and threat modeling for SecureVerse, a zero-trust enterprise security framework.
- **SummerHacks 2026 Backend Lead:** Led backend architecture for the 4-member team that designed F.R.I.D.A.Y, completing the 3-agent mediator engine in under 36 hours.

CERTIFICATIONS & ACCREDITATIONS

- **Advanced Backend Engineering & Distributed Systems Architecture** — Engineering Systems Council (2025)
- **Multi-Agent Systems & LLM Architecture Foundations** — DeepLearning / AI Systems Guild (2026)